STAT 618
Bayesian Statistics

The Final Project

Either solve the following two problems, or replace one or both of them with your own Bayesian problems. Your problems have to involve multiple parameters with prior distributions, data, a real application, and a level of complexity that requires MCMC. Submit your codes, reports, and conclusions by email. Firm deadline: May 4. Solutions will be posted on May 5. You can work by yourself or in groups of two or three. Each group submits one project and shares the same grade.

1. Consider a clinical trial of early treatment of a heart attack. The aim of the study was to compare a new treatment to be given as soon as possible after a heart attack with the standard treatment as the control. The trial resulted in the following count data on the thirty-day mortality rate under each treatment:

Event	New treatment	Control treatment	Total
Survived	150	125	275
Did not survive	13	23	36
Total	163	148	311

Let θ_T and θ_C denote the probabilities of survival in the treatment group and the control group respectively. How does the new treatment compare with the control treatment? Answer this question by analyzing the posterior distribution of $\theta_T = \theta_C$, assuming inde- pendent uniform priors on θ_T and θ_C. Include the following steps in your analysis:

(a) Write down the likelihood of the data.

(b) Derive the (marginal) posterior distributions of θ_T and θ_C.

(c) Simulate draws from the posterior distribution of θ_T/θ_C, and summarize this posterior distribution - describe the posterior density, estimate posterior mean, median, and mode.

(d) Test the null hypothesis $H_0 : \theta_T \leq \theta_C$ (the new treatment offers no improvement) versus the alternative $H_0 : \theta_T \geq 1.2\theta_C$ (at least a 20% increase of the survival rate). Assume that the loss due to the Type I error is twice the loss due to the Type II error.

(e) Clearly state your conclusion.

2. The table below, available on our course web page, displays data from a chemical experiment.

x_1, Reactor temperature (^oC)	x_2, Ratio of H_2 to n-heptane (mole ratio)	x_3, Contact time (sec)	y, Conversion of n-heptane to acetylene (%)
1300	7.5	0.012	49
1300	9	0.012	50.2
1300	11	0.0115	50.5
1300	13.5	0.013	48.5
1300	17	0.0135	47.5
1300	23	0.012	44.5
1200	5.3	0.04	28
1200	7.5	0.038	31.5
1200	11	0.032	34.5
1200	13.5	0.026	35
1200	17	0.034	38
1200	23	0.041	38.5
1100	5.3	0.084	15
1100	7.5	0.098	17
1100	11	0.092	20.5
1100	17	0.086	19.5

A quadratic regression model has been used to explain the outcome y in terms of variables x_1, x_2, x_3. That is, it is assumed that for $i = 1, \ldots, 16$,

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \beta_4 x_{1i} x_{2i} + \beta_5 x_{1i} x_{3i} + \beta_6 x_{2i} x_{3i} + \beta_7 x_{1i}^2 + \beta_8 x_{2i}^2 + \beta_9 x_{3i}^2 + \varepsilon_i,$$

where ε_i are independent "error" terms (noise) having Normal$(0, \sigma^2)$ distribution.

Each regression coefficient β_j has a Normal$(\mu_j, 10^6)$ prior distribution for $j = 0, \ldots, 9$, independently of the other regression coefficients. The hyperparameters $\mu_0, \ldots, \mu_9$ are independently generated from a Normal$(0, 10^6)$ hyper-prior. Recall that the second parameter in WinBUGS' `dnorm(... , ...)` is understood as $1/\sigma^2$. Finally, assume a Uniform(0,100) prior for the error standard deviation σ.

(a) Estimate all the regression coefficients and the error variance under the squared-error loss.

(b) Construct approximate 95% HPD credible sets for $\beta_1, \ldots, \beta_9$ and use them to discuss significance of

  i. main effects x_1, x_2, and x_3;

  ii. interaction effects $x_1 x_2$, $x_1 x_3$, and $x_2 x_3$;

  iii. quadratic terms x_1^2, x_2^2, and x_3^2.

(c) Clearly state your conclusion.